Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
4	*(a)*		$\frac{40}{T} = \cos 36°$ (1) T = 49.4 [N] seen [1] **Alternative:** Resistive force = 50cos36° [1] = 40.5 [N] seen [1] **Alternative:** $\theta = \cos^{-1}\left(\frac{40}{50}\right)$ [1] θ = 36.9° [1]	1	1		2	2	
	(b)	(i)	Horizontal component of tension increased **and consequence i.e.** reference to unbalanced forces (e.g. forward force > than resistive force) Accept greater horizontal resultant force [from dog]		1		1		
		(ii)	ΣF = 49.4 cos20° - 40 [= 6.4 N] [1] $a = \frac{6.4}{35}$ **ecf** on ΣF [1] a = 0.18 [m s^{-2}] [1] N.B. If 50 N used then ΣF = 7 N and a = 0.2 [m s^{-2}]	1	1 1		3	3	

Question			Marking details	Marks available AO1	AO2	AO3	Total	Maths	Prac
	(c)		Rate of doing work = $F_H \times v$ **or** $F_H \times \frac{d}{t}$ [1] accept answers based on work done i.e. work = $Fx \cos\theta$ or $W = F_H x$ 1×[1] for one of: • F_H has increased [v has increased – award mark if no reference made to v or $\frac{d}{t}$ here] ✓ • $\cos\theta$ increased ✓ • as v increases drag increases, so more work is done against drag (per unit distance and time) ✓ Therefore P increased / claim is incorrect [1] accept answer based on work rather than rate of doing work			3	3		
			Question 4 total	**2**	**4**	**3**	**9**	**5**	**0**